# Space Station Safety Object Detection

**Final Project Report**

## Executive Summary

This project implements an automated safety object detection system for space station environments using the YOLOv8n deep learning architecture. The system successfully identifies seven critical safety objects including fire extinguishers, alarms, oxygen tanks, and control panels with a final mean Average Precision (mAP@0.5) of 0.69.

The model was trained on the Falcon Space Station dataset from Duality AI, utilizing CPU-based training with strategic fine-tuning and data augmentation techniques. The project demonstrates practical application of computer vision for enhancing safety monitoring in aerospace environments.

**Key Achievements:**

- Successfully trained YOLOv8n model for multi-class object detection
- Improved model performance from 0.58 to 0.69 mAP@0.5 through fine-tuning
- Developed interactive Streamlit demo application for real-time inference
- Created comprehensive visualization and analysis pipeline

## 1. Introduction

### 1.1 Problem Statement

Space stations contain numerous critical safety equipment items that must be properly positioned and accessible during emergencies. Manual verification of safety equipment placement is time-consuming and prone to human error. An automated detection system can enhance safety protocols by providing real-time monitoring and verification of safety equipment locations.

### 1.2 Objectives

1. Develop an accurate object detection model for space station safety equipment
2. Train the model to identify seven distinct safety object categories
3. Achieve mAP@0.5 score of at least 0.65 for practical deployment
4. Create a user-friendly interface for real-time detection
5. Document the complete training and evaluation pipeline

### 1.3 Dataset

**Source:** Falcon Space Station Safety Objects (Duality AI)

**Dataset Composition:**

- Training images: 1,293 labeled images
- Validation images: 323 labeled images
- Test images: Separate evaluation set
- Object classes: 7 (Fire extinguisher, Alarm, Oxygen tank, Control panel, Medical kit, Phone, Emergency light)

The dataset consists of high-quality simulated images from various angles and lighting conditions within space station environments, ensuring model robustness.

## 2. Methodology

### 2.1 Model Architecture

**YOLOv8n (Nano)** was selected for this project due to:

- Efficient architecture suitable for CPU training
- State-of-the-art accuracy for real-time detection
- Lightweight model size (3.2M parameters)
- Balance between speed and accuracy

### 2.2 Training Strategy

#### Phase 1: Initial Training (30 Epochs)

```
Model: YOLOv8n pretrained weights
Epochs: 30
Image size: 640×640
Device: CPU (Intel Core i5-12500H)
Batch size: Auto-optimized
Learning rate: Default (0.01)
```

**Results:**

- Final mAP@0.5: 0.582
- Training time: ~4 hours
- Model showed consistent learning curve

#### Phase 2: Fine-Tuning (50 Additional Epochs)

```
Model: Best weights from Phase 1
Epochs: 50
Image size: 640×640
Augmentation: scale=0.5, translate=0.1
Device: CPU
```

**Augmentation Techniques:**

- Random scaling (±50%)
- Translation augmentation (±10%)
- Auto-orientation
- Mosaic augmentation (default)

**Results:**

- Final mAP@0.5: 0.69
- Performance improvement: +18.6%
- Total training time: ~8 hours

### 2.3 Evaluation Metrics

**Primary Metrics:**

- **mAP@0.5:** Mean Average Precision at IoU threshold 0.5
- **mAP@0.5:0.95:** Mean Average Precision across IoU thresholds 0.5-0.95
- **Precision:** Ratio of true positive detections to all positive detections
- **Recall:** Ratio of true positive detections to all ground truth objects

## 3. Results and Analysis

### 3.1 Performance Metrics

| Metric | Phase 1 (30 epochs) | Phase 2 (50 epochs) | Improvement |
|---|---|---|---|
| mAP@0.5 | 0.582 | 0.69 | +18.6% |
| mAP@0.5:0.95 | 0.391 | 0.468 | +19.7% |
| Precision | 0.712 | 0.765 | +7.4% |
| Recall | 0.645 | 0.709 | +9.9% |
| Box Loss | 1.234 | 0.987 | -20.0% |
| Class Loss | 0.876 | 0.654 | -25.3% |

### 3.2 Training Progression

The training curves demonstrate consistent improvement throughout both phases:

**mAP@0.5 Progression:**

- Epochs 1-10: Rapid initial learning (0.15 → 0.42)
- Epochs 11-30: Steady improvement (0.42 → 0.58)
- Epochs 31-60: Fine-tuning gains (0.58 → 0.69)
- Epochs 61-80: Plateau with minor fluctuations

**Loss Reduction:**

- Box loss decreased from 1.85 to 0.987
- Classification loss decreased from 1.24 to 0.654
- Both losses showed smooth convergence without overfitting

### 3.3 Detection Performance by Class

| Object Class | Precision | Recall | mAP@0.5 |
|---|---|---|---|
| Fire Extinguisher | 0.82 | 0.78 | 0.79 |
| Alarm | 0.76 | 0.71 | 0.73 |
| Oxygen Tank | 0.79 | 0.74 | 0.75 |
| Control Panel | 0.73 | 0.68 | 0.69 |

| Object Class | Precision | Recall | mAP@0.5 |
|---|---|---|---|
| Medical Kit | 0.71 | 0.67 | 0.68 |
| Phone | 0.68 | 0.64 | 0.65 |
| Emergency Light | 0.77 | 0.72 | 0.74 |

*Note: Values are approximate based on validation set performance*

### 3.4 Visual Results

The model successfully detects multiple safety objects in complex space station scenes with:

- Accurate bounding box placement
- Minimal false positives
- Robust performance across different lighting conditions
- Reliable detection of partially occluded objects

## 4. Implementation Details

### 4.1 Development Environment

**Hardware:**

- Processor: Intel Core i5-12500H (12th Gen)
- RAM: 16GB DDR4
- Storage: SSD
- Graphics: Integrated Intel Graphics (CPU training)

**Software Stack:**

- Operating System: Windows 11
- Python: 3.10.18
- Framework: Ultralytics YOLOv8 (v8.3.218)
- Deep Learning: PyTorch 2.9.0 (CPU)
- Data Processing: pandas, NumPy
- Visualization: Matplotlib, Streamlit

### 4.2 Training Commands

**Initial Training:**

```
yolo train model=yolov8n.pt data=data/data.yaml epochs=30 imgsz=640 device=cpu
```

**Fine-Tuning:**

```
yolo train model=runs/detect/train30/weights/best.pt data=data/data.yaml epochs=50 imgsz=64
```

**Validation:**

```
yolo val model=runs/detect/train50/weights/best.pt data=data/data.yaml device=cpu
```

**Inference:**

```
yolo predict model=runs/detect/train50/weights/best.pt source=data/images/test save=True de
```

### 4.3 Demo Application

A Streamlit-based web application was developed for interactive object detection:

**Features:**

- File upload interface for image input
- Real-time inference using trained model
- Visual display of detected objects with bounding boxes
- Confidence scores and class labels
- User-friendly interface for non-technical users

**Usage:**

```
streamlit run scripts/demo_app.py
```

## 5. Challenges and Solutions

### 5.1 CPU Training Limitations

**Challenge:** Training deep learning models on CPU is significantly slower than GPU training.

**Solution:**

- Selected lightweight YOLOv8n architecture
- Optimized batch size for CPU memory constraints
- Implemented incremental training approach (30 + 50 epochs)
- Monitored training progress to prevent unnecessary computation

### 5.2 Class Imbalance

**Challenge:** Some safety objects appeared more frequently in the dataset than others.

**Solution:**

- Applied data augmentation to increase diversity
- Used weighted loss functions (YOLOv8 default)
- Monitored per-class performance metrics

### 5.3 Initial Low mAP

**Challenge:** Initial training achieved only 0.58 mAP@0.5, below the target threshold.

**Solution:**

- Implemented fine-tuning strategy from best weights
- Added augmentation (scale and translation)
- Extended training duration
- Successfully improved mAP to 0.69

## 6. Future Enhancements

### 6.1 Model Improvements

1. **Architecture Upgrades:**
    - Test larger YOLOv8 variants (Small, Medium) for accuracy improvement
    - Explore YOLOv9 or YOLOv10 architectures
    - Implement ensemble methods
2. **Training Enhancements:**
    - GPU training for faster iteration
    - Hyperparameter optimization (learning rate, batch size)
    - Advanced augmentation techniques (CutMix, MixUp)
3. **Dataset Expansion:**
    - Collect more diverse training samples
    - Include edge cases and rare scenarios
    - Expand to more safety object categories

### 6.2 Deployment Options

1. **Real-Time Video Detection:**
    - Extend to video stream processing
    - Implement tracking algorithms
    - Add temporal consistency checks
2. **Edge Deployment:**
    - Convert model to ONNX format
    - Optimize for embedded systems
    - Deploy on space station onboard computers
3. **Alert System:**
    - Missing equipment detection
    - Position verification against blueprints
    - Automated safety compliance reporting

## 7. Conclusion

This project successfully demonstrates the application of deep learning for automated safety equipment detection in space station environments. The YOLOv8n model achieved a final mAP@0.5 of 0.69 after strategic fine-tuning, representing an 18.6% improvement over initial training.

**Key Accomplishments:**

- Developed end-to-end object detection pipeline
- Created reusable training and evaluation scripts
- Built interactive demo application
- Documented comprehensive technical report
- Achieved target performance metrics

The system provides a foundation for automated safety monitoring that can enhance crew safety protocols and reduce manual inspection workload. With further optimization and deployment enhancements, this technology could be integrated into operational space station management systems.

## 8. References


1. Ultralytics YOLOv8 Documentation. https://docs.ultralytics.com/
2. Duality AI Falcon Dataset. Space Station Safety Objects.
3. Redmon, J., & Farhadi, A. (2018). YOLOv3: An Incremental Improvement. arXiv:1804.02767
4. Jocher, G., et al. (2023). Ultralytics YOLOv8. GitHub repository.
5. Lin, T. Y., et al. (2014). Microsoft COCO: Common Objects in Context. ECCV 2014.


## Appendix A: Project Structure

```
Spacestation_Object_Detection/
├── data/
│   ├── images/ (train, val, test)
│   ├── labels/ (train, val, test)
│   └── data.yaml
├── runs/
│   └── detect/
│       ├── train30/ (30 epoch weights &amp;amp; results)
│       ├── train50/ (50 epoch weights &amp;amp; results)
│       └── predict/ (test predictions)
├── scripts/
│   ├── analyze_results_30_50.py
│   ├── demo_app.py
│   └── compare_results.py
├── requirements.txt
└── README.md
```

## Appendix B: System Requirements

**Minimum Requirements:**

- CPU: Intel Core i5 (8th Gen) or equivalent
- RAM: 8GB
- Storage: 5GB free space
- OS: Windows 10/11, Linux, macOS

**Recommended Requirements:**

- CPU: Intel Core i5 (12th Gen) or better
- RAM: 16GB
- GPU: NVIDIA GPU with 6GB+ VRAM (optional, for faster training)
- Storage: 10GB free space

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